DECISION INTELLIGENCE CASE STUDY

Which store gets the last unit?

Decision-focused and risk-aware stock allocation for a multi-store retailer, and an honest test of whether the advanced methods are worth it

DATA

Walmart M5 sales (public) as a demand proxy

ECONOMICS

Synthetic monetary units (MU), not real costs

EVIDENCE

12 test weeks · 2 demand regimes · 3 seeds

THE ANSWER FIRST

Three findings in one slide

−133 MU

per week (9.3% lower cost) from moving off an equal split to any demand-aware rule

95% interval 114 to 155 MU · regular demand

Tie

Decision-focused learning and robust optimisation did not beat a simple proportional rule on regular demand

every model's 95% interval vs the rule crosses zero

Unstable

On intermittent demand, forecast → optimise won clearly on one seed, but not when retrained

best policy by seed: seed 42: Forecast → robust; seed 101: Equal split; seed 202: Equal split

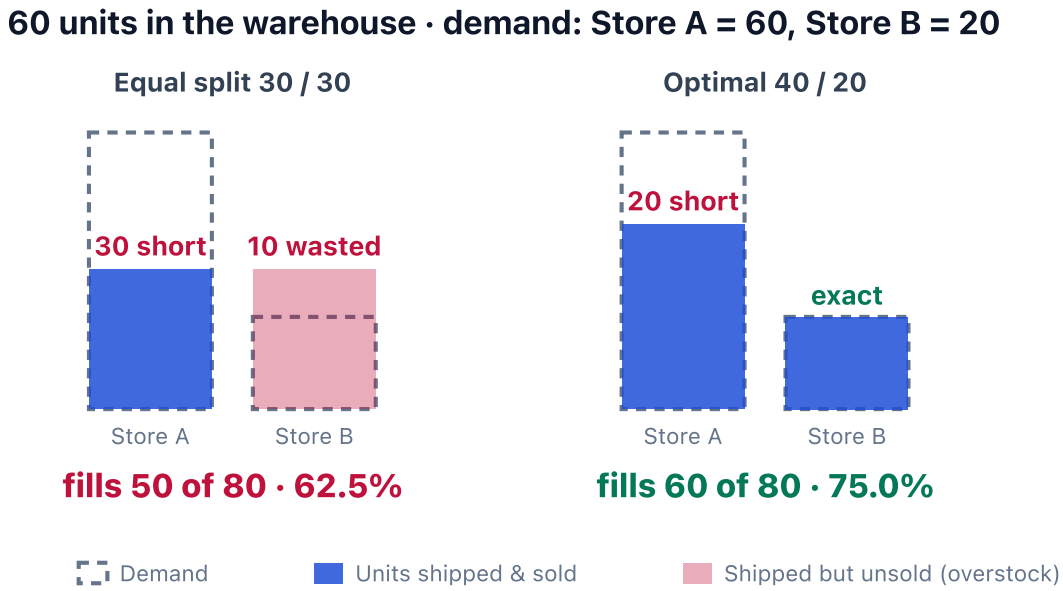
What I would tell a business owner: get the allocation logic right first, it is where the money is. Add advanced learning only where a controlled test shows it pays. I built the evidence system that can tell the difference.

1 · PROBLEM

Same stock, different outcome: allocation is the decision

- A warehouse rarely has enough to cover every store. Every unit sent to one store is a unit another store doesn't get.
- The decision is a whole number of units per store, under warehouse supply, shelf space and per-item caps.
- In this tiny example, the same 60 units fill **20% more demand** just by sending them where they will sell.

> This exact example is one of the project's automated tests: the optimiser must find 40 / 20.

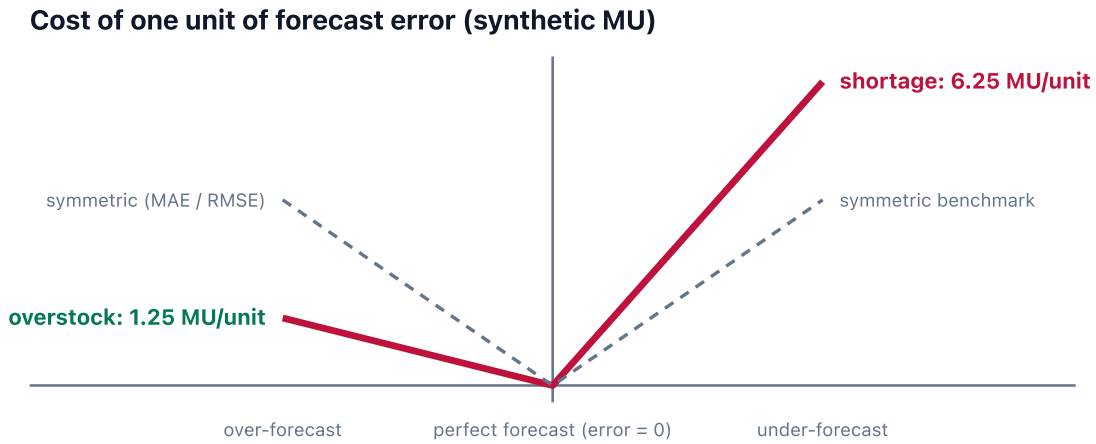


1 · PROBLEM

A better forecast score doesn't guarantee a better decision

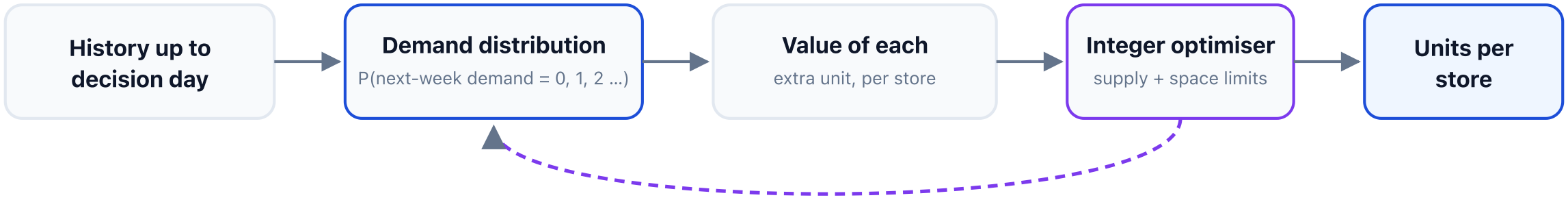
- In this setup, running out costs **6.25 MU** per unit; overstocking costs **1.25 MU**. Roughly **5 : 1**.
- Accuracy metrics such as MAE treat both errors as equally bad, so a model tuned for accuracy is tuned for the wrong goal.
- The quantity to measure is the **cost of the decision**, compared with the best decision possible in hindsight (**regret**).

> Costs are synthetic. In a real deployment they would come from margin, holding and freight data.



2 · APPROACH

From history to shipments, with the optimiser in the training loop



DFL training loop: optimiser penalises decision regret (SPO+ loss) → updates forecaster

Forecast → optimise

Train the forecaster for calibrated probabilities, then optimise. The standard approach and a fair baseline.

Decision-focused learning (DFL)

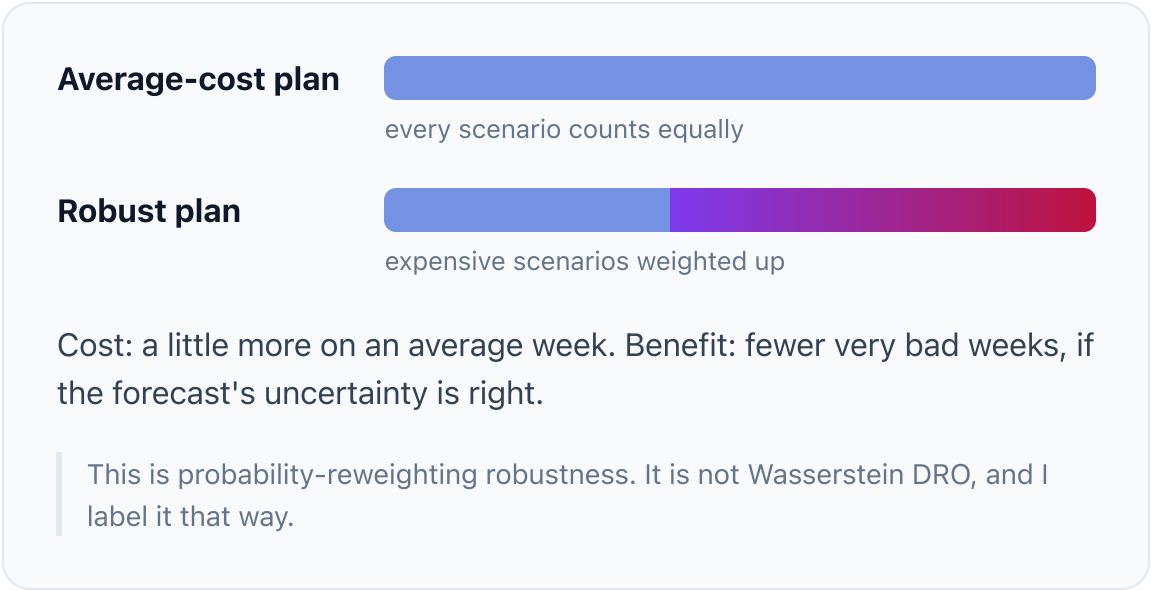
During training, the optimiser scores each forecast by the **cost of the decision it led to** (SPO+ loss), and the model learns from that.

Same model, same budget

Both versions share architecture, initial weights, data order and epochs. Only the training signal differs.

Risk-aware allocation: plan for the bad weeks, not only the average

- Simulate many possible next weeks from the forecast, keeping stores' tendency to move together.
- The robust optimiser gives extra weight to the most expensive scenarios. Mathematically, this equals minimising the average cost of the worst tail of outcomes (CVaR).
- Two dials: how far into the tail (α) and how much to care about it (risk weight). Both chosen on validation data only: α = 0.75, weight = 0.25.
- Setting the risk weight to zero gives back the ordinary average-cost plan, a built-in sanity check.



Cost: a little more on an average week. Benefit: fewer very bad weeks, if the forecast's uncertainty is right.

> This is probability-reweighting robustness. It is not Wasserstein DRO, and I label it that way.

3 · ANALYSIS

How I made the comparison fair



Time only moves forward: the model never sees the future it is tested on.

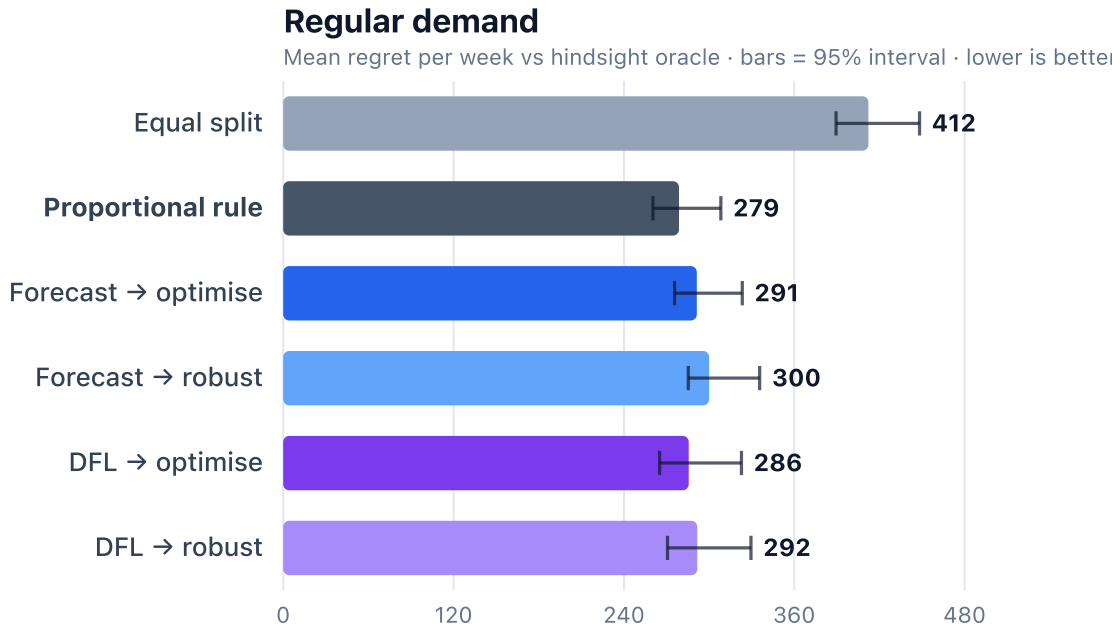
Same conditions
Every policy faces the same weeks, stock, supply, shelf limits and costs.

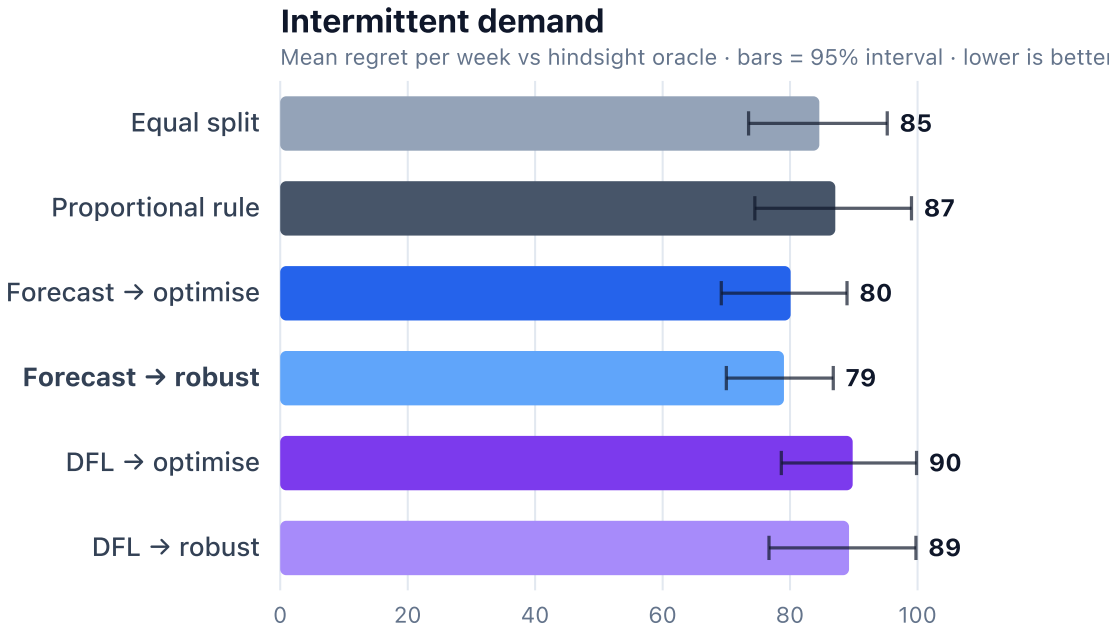
Two regimes
5 stores × 10 household items each: regular sellers and intermittent sellers.

Strong baselines
Equal split and a proportional rule (last 8 weeks of sales), as a retailer would use.

Checked independently
All 288 policy-week allocations re-verified: feasible, costs recomputed, no negative regret.

Results: big win over equal split; models tie with a simple rule

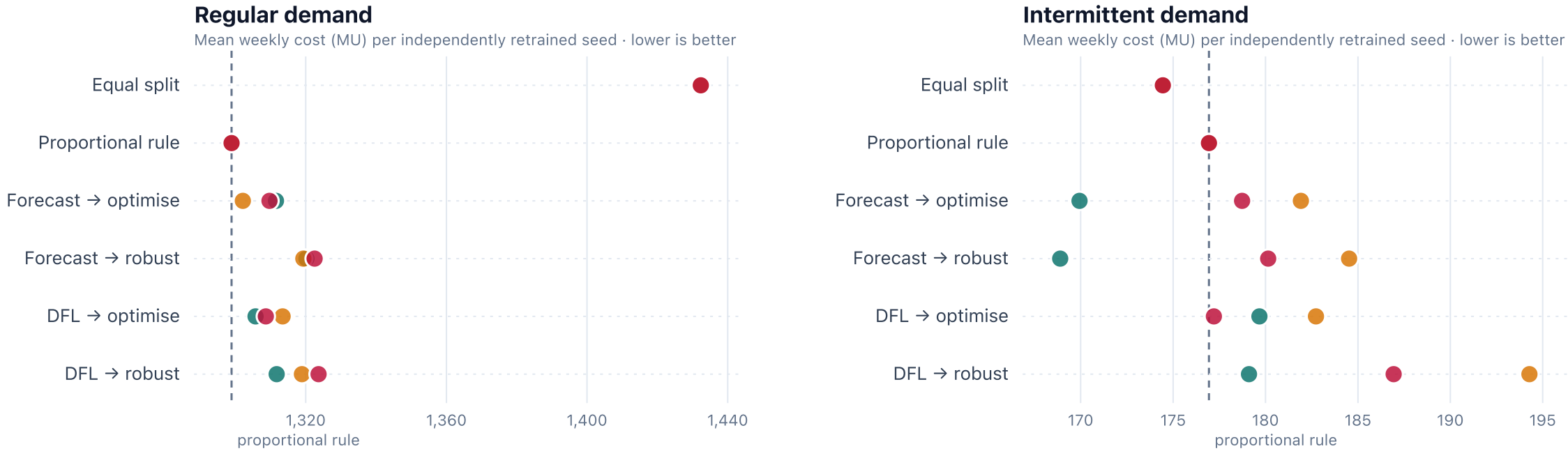




Read it this way: regular demand, equal split is clearly worst and the rest overlap. Intermittent demand, forecast → optimise is the only policy clearly better than the proportional rule on the primary seed (-7.0 MU, interval [-11.4, -2.5]). Even the best policy is ~279 MU from perfect hindsight, which is the limit of predictability.

3 · ANALYSIS

Do the rankings survive retraining? Not reliably



Best policy by seed. Regular: seed 42: Proportional rule; seed 101: Proportional rule; seed 202: Proportional rule. Intermittent: seed 42: Forecast → robust; seed 101: Equal split; seed 202: Equal split. A result that changes with the random seed is not yet a basis for a business decision.

Live demo: the decision tool

1. **Page A**: data source badge, synthetic costs, time split
2. **Page C**: units per store, supply and shelf limits respected
3. **Page D**: cut warehouse supply to 0.6× and watch stock get rationed
4. **Page D**: move the risk dial; at α = 0 the risk cost equals the average cost
5. **Page E**: paired results and CSV export

`localhost:8511`

Runs fully offline on a laptop

- Models pre-trained; optimisation runs live
- Nominal solve ≈ 0.02 s, robust solve ≈ 0.1–0.15 s per week
- All 15 page × mode views checked in an automated browser test

> Backup if anything fails: this deck plus the PDF.

5 · IMPACT

What the evidence supports, and what it doesn't

Supported

- Demand-aware allocation beats an equal split by about **133 MU/week (9.3%)** in this setup, with a clear interval.
- An open-source integer optimiser handles the constraints in a fraction of a second per week.
- The evaluation harness catches lucky results: paired intervals, retrained seeds, independent recomputation.

Not supported (yet)

- That decision-focused learning or robust optimisation beats a simple rule on this data.
- Any real saving in RM: costs are synthetic and sales stand in for demand.
- Performance at network scale: this is 5 stores × 10 items per regime.

Recommended next step: run the proportional rule plus optimiser as the baseline, and treat DFL / robust optimisation as challengers that must win a pre-registered shadow test.

5 · IMPACT

Limitations and what I would do next

LIMITATION	WHY IT MATTERS	NEXT STEP
Sales stand in for demand	Stockouts hide true demand, so costs are understated	Use on-hand stock records; censored-demand model
Synthetic costs	Rankings can change with the real cost ratio	Calibrate from margin, holding and freight data
12 test weeks, small cohorts	Too few to separate methods 5–10 MU apart	Longer, untouched holdout; more items and stores
Test period seen before the fixes	This is a corrected re-run, not a fresh confirmatory test	Pre-register the next evaluation on new weeks
Single-week decisions	Real stock carries over between weeks	Extend the tested multi-week simulator to the main comparison

5 · IMPACT

From prototype to pilot: a low-risk path

Phase 1 · Shadow

Weeks 1–8

Connect real stock, sales and cost data. Generate recommendations alongside planners; ship nothing. Measure regret vs current orders.

Go if: clear, significant gain vs current practice

Phase 2 · Pilot

Weeks 9–16

Small store cluster and a focused item set. Planner can override; every override is logged and reviewed.

Go if: fewer stockouts, no rise in overstock

Phase 3 · Scale

After pilot review

Solve items in parallel (each item's warehouse limit is separate), then reconcile shelf space. Monitor drift.

Challenger models promoted only on evidence

SUMMARY

Get the decision right first. Prove complexity before you ship it.

BUILT

End-to-end pipeline, integer optimiser, decision-focused and robust variants, offline dashboard

VERIFIED

47 automated tests, 288 allocations recomputed, leakage checks, 3 retrained seeds

FOUND

~9% gain from allocation logic; advanced methods not yet proven on this data

Thank you. Happy to take questions or go deeper on any part.

BACKUP

B1 · The optimisation model

COST FOR ONE WEEK (PER STORE S, ITEM P)

$$\text{cost} = \Sigma\, b \cdot \text{shortage}_{sp} + h \cdot \text{leftover}_{sp} + t \cdot \text{shipped}_{sp}$$

$$\text{shortage} = \max(D - I - x, 0) \quad \text{leftover} = \max(I + x - D, 0)$$

CONSTRAINTS

$$\Sigma_s\, x_{sp} \le W_p \text{ (warehouse)} \quad \Sigma_p\, (I_{sp} + x_{sp}) \le C_s \text{ (store space)} \quad 0 \le x_{sp} \le K_{sp}, \text{ integer}$$

VALUE OF THE K-TH EXTRA UNIT (EXACT IDENTITY, UNIT-TESTED)

$c[s,k] = t - b$ if demand $\ge I + k$ (it prevents a shortage), else $t + h$ (it is left over)

expected: $\hat{c}[s,k] = t + h - (b + h) \cdot P(D \ge I + k)$

> b = 6.0, h = 1.0, t = 0.25 MU per unit (synthetic). Solver: HiGHS (open source). Fulfilment is recomputed after solving as min(demand, stock).

BACKUP

B2 · Decision-focused learning details

SPO+ LOSS (ELMACHTOUB & GRIGAS), USED VIA PYEPO

$$L(\hat{c}, c) = \max_{w \in S} (c - 2\hat{c})^T w + 2\hat{c}^T w^*(c) - c^T w^*(c)$$

TRAINING OBJECTIVE

$$\text{loss} = \text{NLL}(\text{demand distribution}) + \lambda \cdot \text{SPO+}, \ \lambda \text{ chosen on validation from } \{0, 0.5, 1, 2\}$$

- Training uses a fixed set of constraints per item, so SPO+'s assumptions hold. Real weekly limits are applied at decision time.
- Gradient check: SPO+ gradient is finite and non-zero; $\lambda = 0$ vs $\lambda > 0$ produce different updates.
- Validation regret is recomputed after restoring the best checkpoint and calibration.

B3 · Why the robust model equals CVaR

AMBIGUITY SET OVER N SCENARIOS WITH BASE PROBABILITIES P

$$P_\alpha = \{\, w : \Sigma\, w_j = 1,\ 0 \le w_j \le p_j / (1-\alpha) \,\}$$

$$\max_{w \in P_\alpha} \Sigma\, w_j\, L_j(x) = \mathrm{CVaR}_\alpha(L(x))$$

OBJECTIVE

$$\min\ (1-\rho)\cdot E[L(x)] + \rho \cdot \mathrm{CVaR}_\alpha(L(x))$$

- Solved exactly as one integer program with the standard CVaR linearisation (η, z_j).
- Checked against brute-force enumeration on small cases; $\alpha = 0$ or $\rho = 0$ recovers the average-cost plan.
- Scenarios share random numbers across policies and keep cross-store dependence learned from training weeks only.

BACKUP

B4 · Full results, primary seed (MU per week)

Regular demand

POLICY	COST	REGRET	FILL	WORST-10% COST
Equal split	1,432.3	412.2	71.2%	1,843.0
Proportional rule	1,298.9	278.8	73.9%	1,711.7
Forecast → optimise	1,311.5	291.4	73.7%	1,688.0
Forecast → robust	1,320.2	300.0	73.5%	1,698.4
DFL → optimise	1,305.7	285.6	73.8%	1,674.7
DFL → robust	1,311.7	291.6	73.7%	1,684.0
Hindsight oracle	1,020.2	0.0	79.2%	1,536.6

vs proportional rule

POLICY	VS PROPORTIONAL	95% INTERVAL	VERDICT
Forecast → optimise	+12.6	[-8.8, +48.7]	within noise
Forecast → robust	+21.3	[-6.7, +61.0]	within noise
DFL → optimise	+6.8	[-14.9, +44.8]	within noise
DFL → robust	+12.8	[-16.7, +54.3]	within noise

Intermittent demand

POLICY	COST	REGRET	FILL	WORST-10% COST
Equal split	174.4	84.6	56.2%	269.2
Proportional rule	176.9	87.1	55.3%	274.4
Forecast → optimise	169.9	80.1	57.3%	261.3
Forecast → robust	168.9	79.0	56.4%	268.0
DFL → optimise	179.7	89.8	53.6%	266.6
DFL → robust	179.1	89.3	53.3%	265.1
Hindsight oracle	89.9	0.0	73.7%	160.8

vs proportional rule

POLICY	VS PROPORTIONAL	95% INTERVAL	VERDICT
Forecast → optimise	-7.0	[-11.4, -2.5]	clearly better
Forecast → robust	-8.1	[-19.2, +2.1]	within noise
DFL → optimise	+2.7	[-3.1, +9.5]	within noise
DFL → robust	+2.2	[-7.0, +11.0]	within noise

> Worst-10% cost averages roughly the single most expensive of 12 test weeks, so it is noisy. Intervals: paired moving-block bootstrap over weeks (2,000 resamples).